

Logarithms

Question1

$\cosh(\log 4)$ is equal to

AP EAPCET 2024 - 23th May Morning Shift

Options:

A. $\frac{8}{17}$

B. $\frac{17}{8}$

C. 0

D. $\frac{9}{8}$

Answer: B

Solution:

To find $\cosh(\log 4)$, we start by using the identity for the hyperbolic cosine function:

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Substituting $x = \log 4$ into the formula, we have:

$$\cosh(\log 4) = \frac{e^{\log 4} + e^{-\log 4}}{2}$$

We know that:

$$e^{\log 4} = 4 \quad \text{and} \quad e^{-\log 4} = \frac{1}{4}$$

Substitute these values back into the equation:

$$\cosh(\log 4) = \frac{4 + \frac{1}{4}}{2} = \frac{\frac{16}{4} + \frac{1}{4}}{2}$$

Simplify the expression:

$$= \frac{\frac{17}{4}}{2} = \frac{17}{8}$$

Therefore, $\cosh(\log 4) = \frac{17}{8}$.

Question2

$$\cosh 1 + 2 =$$

AP EAPCET 2024 - 18th May Morning Shift

Options:

A. $\log(2 + \sqrt{3})$

B. $\log(2 + \sqrt{5})$

C. $\log(2 - \sqrt{5})$

D. $\log(2 + \sqrt{2})$

Answer: A

Solution:

To solve this problem, let's use the relationship for the inverse hyperbolic cosine function:

$$\cosh^{-1} x = \log \left(x + \sqrt{x^2 - 1} \right)$$

Applying this to $\cosh^{-1} 2$, we get:

$$\cosh^{-1} 2 = \log \left(2 + \sqrt{2^2 - 1} \right)$$

Simplifying under the square root:

$$\sqrt{2^2 - 1} = \sqrt{4 - 1} = \sqrt{3}$$

So, the expression becomes:

$$\cosh^{-1} 2 = \log(2 + \sqrt{3})$$

Question3

$$4^x - 3^{x-\frac{1}{2}} = 3^{x+\frac{1}{2}} - 2^{2x-1} \Rightarrow x =$$

AP EAPCET 2022 - 5th July Morning Shift

Options:

A. $\frac{5}{2}$

B. $\frac{1}{2}$

C. $\frac{3}{2}$

D. $\frac{7}{2}$

Answer: C

Solution:

Given,

$$\begin{aligned}4^x - 3^{x-\frac{1}{2}} &= 3^{x+\frac{1}{2}} - 2^{2x-1} \\ \Rightarrow (2)^{2x} + 2^{2x-1} &= 3^{x+\frac{1}{2}} + 3^{x-\frac{1}{2}} \\ \Rightarrow 2^{2x} + \frac{2^{2x}}{2} &= 3^x \cdot \sqrt{3} + \frac{3^x}{\sqrt{3}} \\ \Rightarrow 2^{2x} \left[1 + \frac{1}{2} \right] &= 3^x \left[\sqrt{3} + \frac{1}{\sqrt{3}} \right] \\ \Rightarrow 2^{2x} \times \frac{3}{2} &= 3^x \times \frac{4}{\sqrt{3}} \\ \Rightarrow 2^{2x}(3\sqrt{3}) &= 3^x(8) \\ \Rightarrow 2^{2x}(3)^{\frac{3}{2}} &= 3^x \times (2)^3\end{aligned}$$

Comparing power of 2, $2x = 3$

and comparing power of 3, $x = \frac{3}{2}$

Both equations given us $x = \frac{3}{2}$.

Question4

$$\left\{ x \in R / \frac{\sqrt{|x|^2 - 2|x| - 8}}{\log(2 - x - x^2)} \text{ is a real number} \right\} =$$

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $(-\infty, -4] \cup [4, \infty)$

B. ϕ

C. $(-1, 2)$

D. $(-\infty, -4] \cup (-1, 2) \cup [4, \infty)$

Answer: B

Solution:

$$\begin{aligned} & \left\{ x \in R / \frac{\sqrt{|x|^2 - 2|x| - 8}}{\log(2 - x - x^2)} \text{ is a real number} \right\} \\ & \left\{ x \in R / \frac{\sqrt{x^2 - 2x - 8}}{\log(2 + x)(1 - x)} \text{ is a real number} \right\} \\ & = \left\{ x \in R / \frac{\sqrt{(x - 4)(x + 2)}}{\log\{(2 + x)(1 - x)\}} \text{ is a real number} \right\} \\ & = \phi \end{aligned}$$

Question5

If $4 + 6(e^{2x} + 1) \tanh x = 11 \cosh x + 11 \sinh x$ then $x =$

AP EAPCET 2022 - 4th July Evening Shift

Options:

A. $\log_{10}$

B. $\log 4$

C. $\log 5$

D. $\log 2$

Answer: D

Solution:

Given,

$$4 + 6(e^{2x} + 1) \tanh x = 11 \cosh x + 11 \sinh x$$

$$\Rightarrow 4 + 6(e^{2x} + 1) \cdot \frac{(e^{2x} - 1)}{(e^{2x} + 1)}$$

$$= 11 \left\{ \frac{e^{2x} + 1}{2e^x} + \frac{e^{2x} - 1}{2e^x} \right\}$$

$$= 11 \left\{ \frac{2e^{2x}}{2e^x} \right\} = 11e^x$$

$$\Rightarrow 4 + 6(e^{2x} - 1) = 11e^x$$

Let $e^x = P$

$$\Rightarrow 4 + 6(P^2 - 1) = 11P$$

$$\Rightarrow 6P^2 - 11P - 2 = 0$$

$$\Rightarrow 6P^2 - 12P + P - 2 = 0$$

$$\Rightarrow (P - 2)(6P + 1) = 0$$

$$\Rightarrow P = 2 \text{ and } \frac{-1}{6}$$

Thus, $e^x = 2$

$$\Rightarrow x = \log_e 2$$

Question6

If $4^x - 3^{x-1/2} = 3^{x+1/2} - 2^{2x-1}$, then the value of x is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. $7/2$

B. $5/2$

C. $1/2$

D. $3/2$

Answer: D

Solution:

$$4^x - 3^{x-\frac{1}{2}} = 3^{x+\frac{1}{2}} - 2^{2x-1}$$

$$\Rightarrow 4^x - 3^x \cdot 3^{-\frac{1}{2}} = 3^x \cdot 3^{\frac{1}{2}} - 2^{2x} \cdot 2^{-1}$$

$$\Rightarrow 4^x - 3^x \cdot \frac{1}{\sqrt{3}} = \sqrt{3} \cdot 3^x - \frac{4^x}{2}$$

$$\Rightarrow 4^x + \frac{4^x}{2} = \sqrt{3} \cdot 3^x + \frac{1}{\sqrt{3}} \cdot 3^x$$

$$\Rightarrow 4^x \left(1 + \frac{1}{2}\right) = 3^x \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow \frac{3}{2} \cdot 4^x = \frac{4}{\sqrt{3}} \cdot 3^x$$

$$\Rightarrow \frac{4^x}{3^x} = \frac{2 \cdot 4}{3\sqrt{3}}$$

$$\Rightarrow \left(\frac{4}{3}\right)^x = \frac{4^{\frac{1}{2}} \cdot 4^1}{3^{1\frac{1}{3^2}}} = \left(\frac{4}{3}\right)^{\frac{1}{2}+1} = \left(\frac{4}{3}\right)^{\frac{3}{2}}$$

$$\Rightarrow x = \frac{3}{2}$$
